MQTT – Current Sensor – Module

Summary:

Module for the ESP32 S3 based modular controller board.
Module is designed to run 3 x 4–20mA Sensors, and read them at a resolution of 15 bit.

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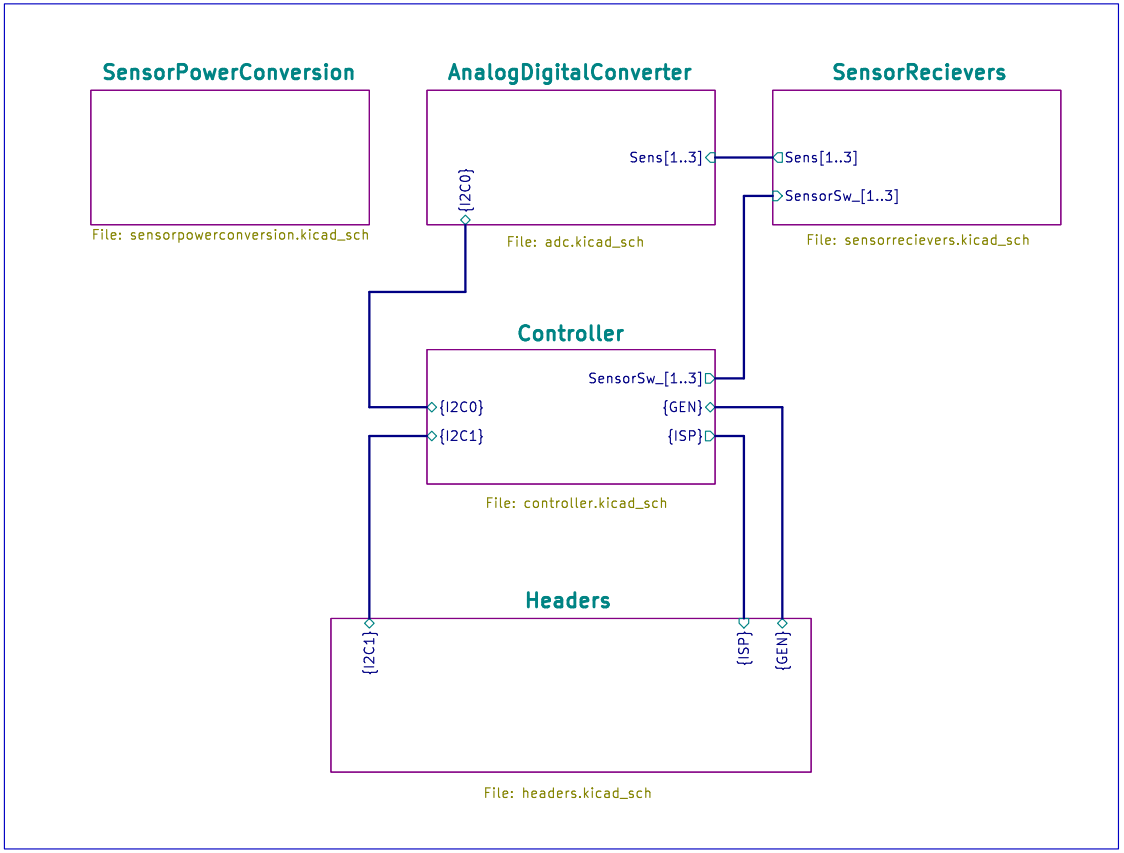
Page 2: Controller
Page 3: Headers
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Non BOM Items

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Additional BOM Items

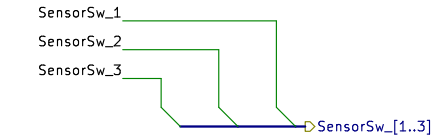
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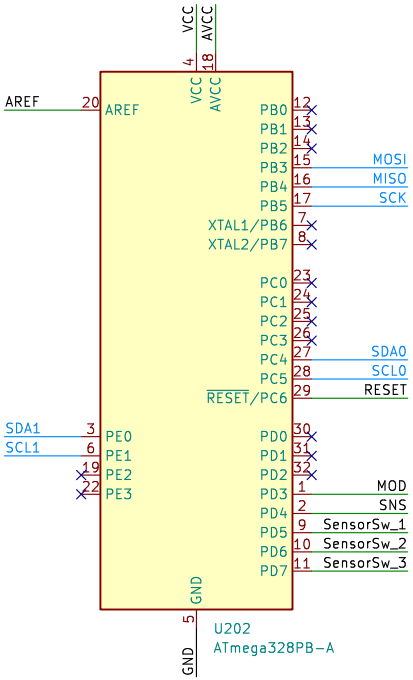


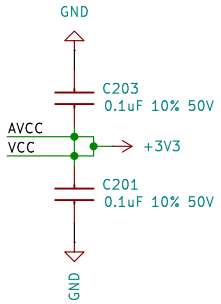
Sheet: /
File: MQTT–Module–CurrentSensors.kicad_sch

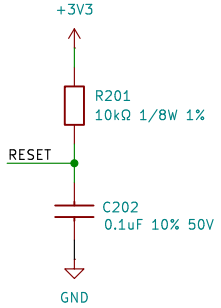
Title: MQTT – Current Sensor – Module

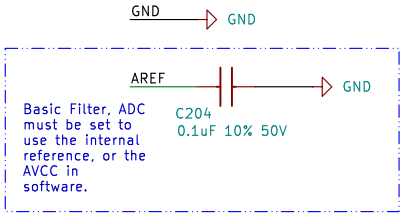
Size: A4	Date: 2026–10–09	**Rev: 1.1**
KiCad E.D.A. 9.0.6		Id: 1/6

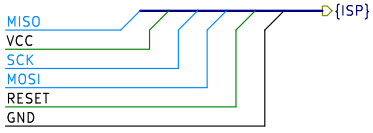


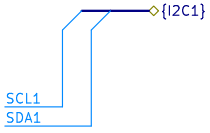


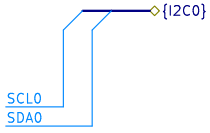


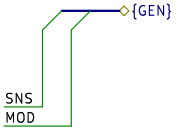










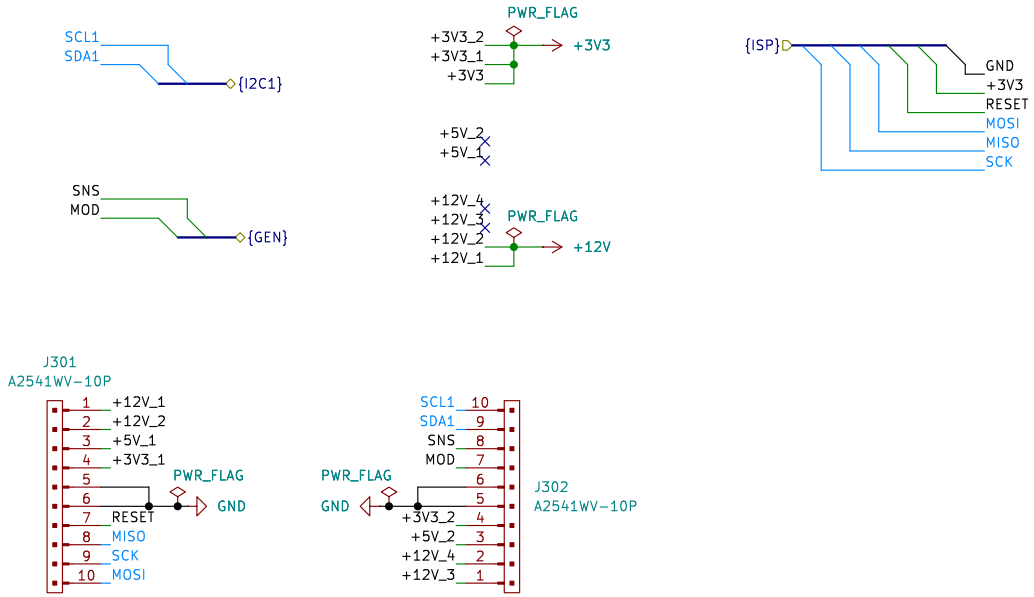


Summary:
– ATMega328PB–A Based control with dual dedicated SDA SCL.

Sheet: /Controller/
File: controller.kicad_sch

Title: Controller

Size: A4	Date: 2026-10-09	**Rev: 1.1**
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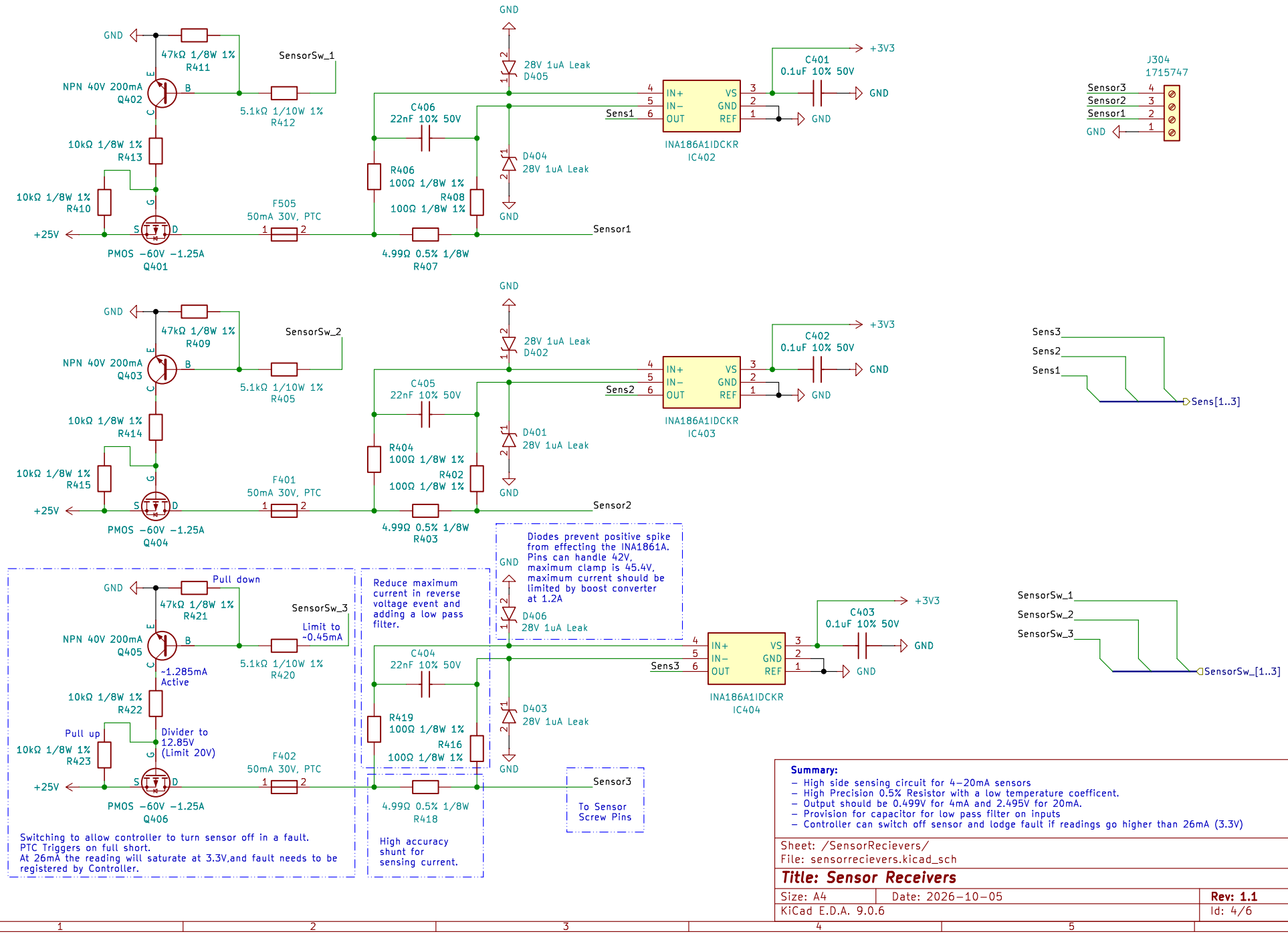


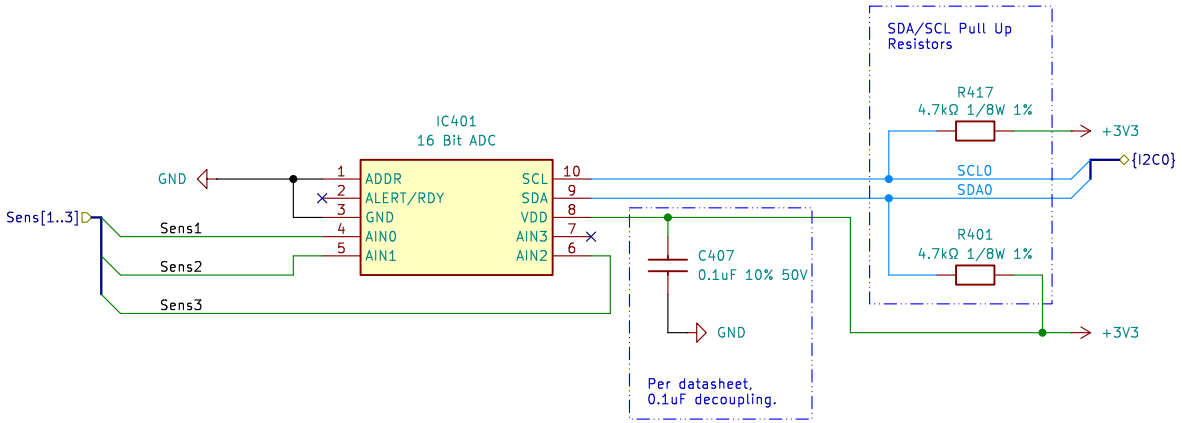
Summary:
- Headers to connect to main board. DNP as they will be hand soldered on.
- ISP Header for connection to MEGA ISP programmer.

Sheet: /Headers/
File: headers.kicad_sch

Title: Headers

Size: A4	Date: 2026-10-05	**Rev: 1.1**
KiCad E.D.A. 9.0.6		Id: 3/6


GND
47kΩ 1/8W 1%
R411
SensorSw_1
NPN 40V 200mA
Q402
5.1kΩ 1/10W 1%
R412
10kΩ 1/8W 1%
R413
10kΩ 1/8W 1%
R410
F505
50mA 30V, PTC
+25V
PMOS −60V −1.25A
Q401
C406
22nF 10% 50V
R406
100Ω 1/8W 1%
R408
100Ω 1/8W 1%
4.99Ω 0.5% 1/8W
R407
Sensor1
28V 1uA Leak
D405
D404
28V 1uA Leak
Sens1
INA186A1IDCKR
IC402
C401
0.1uF 10% 50V
+3V3
J304
1715747
Sensor3
Sensor2
Sensor1
GND
47kΩ 1/8W 1%
R409
SensorSw_2
NPN 40V 200mA
Q403
5.1kΩ 1/10W 1%
R405
10kΩ 1/8W 1%
R414
10kΩ 1/8W 1%
R415
F401
50mA 30V, PTC
+25V
PMOS −60V −1.25A
Q404
C405
22nF 10% 50V
R404
100Ω 1/8W 1%
R402
100Ω 1/8W 1%
4.99Ω 0.5% 1/8W
R403
Sensor2
28V 1uA Leak
D402
D401
28V 1uA Leak
Sens2
INA186A1IDCKR
IC403
C402
0.1uF 10% 50V
+3V3
Sens3
Sens2
Sens1
Sens[1..3]
Diodes prevent positive spike from effecting the INA1861A. Pins can handle 42V, maximum clamp is 45.4V, maximum current should be limited by boost converter at 1.2A
GND
47kΩ 1/8W 1%
R421
Pull down
SensorSw_3
Limit to ~0.45mA
NPN 40V 200mA
Q405
~1.285mA Active
5.1kΩ 1/10W 1%
R420
10kΩ 1/8W 1%
R422
Pull up
10kΩ 1/8W 1%
R423
Divider to 12.85V (Limit 20V)
F402
50mA 30V, PTC
+25V
PMOS −60V −1.25A
Q406
Switching to allow controller to turn sensor off in a fault.
PTC Triggers on full short.
At 26mA the reading will saturate at 3.3V,and fault needs to be registered by Controller.
Reduce maximum current in reverse voltage event and adding a low pass filter.
C404
22nF 10% 50V
R419
100Ω 1/8W 1%
R416
100Ω 1/8W 1%
4.99Ω 0.5% 1/8W
R418
High accuracy shunt for sensing current.
Sensor3
To Sensor Screw Pins
D406
28V 1uA Leak
D403
28V 1uA Leak
Sens3
INA186A1IDCKR
IC404
C403
0.1uF 10% 50V
+3V3
SensorSw_1
SensorSw_2
SensorSw_3
SensorSw_[1..3]
Summary:
– High side sensing circuit for 4–20mA sensors
– High Precision 0.5% Resistor with a low temperature coefficent.
– Output should be 0.499V for 4mA and 2.495V for 20mA.
– Provision for capacitor for low pass filter on inputs
– Controller can switch off sensor and lodge fault if readings go higher than 26mA (3.3V)
Sheet: /SensorRecievers/
File: sensorrecievers.kicad_sch
Title: Sensor Receivers
Size: A4
Date: 2026-10-05
Rev: 1.1
KiCad E.D.A. 9.0.6
Id: 4/6

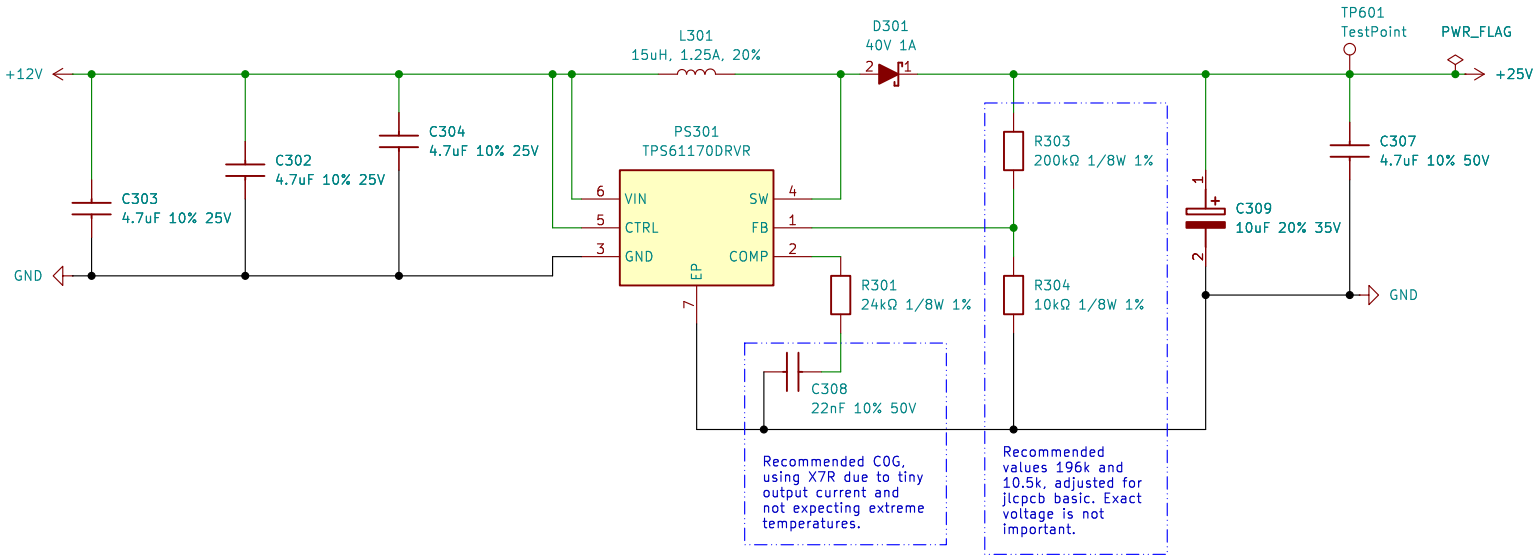


Summary:
– ADS1115 configured to use base address (0x48).

Sheet: /AnalogDigitalConverter/
File: adc.kicad_sch

Title: Analog Digital Converter

Size: A4	Date: 2026-10-09	**Rev: 1.1**
KiCad E.D.A. 9.0.6		Id: 5/6



Summary:
- Texas Instruments generated Boost Converter, 12V to ~25.7V
- Used a shielded inductor for minimal interference.

Sheet: /SensorPowerConversion/		
File: sensorpowerconversion.kicad_sch		
Title: Sensor Power Conversion		
Size: A4	Date: 2026-10-06	**Rev: 1.1**
KiCad E.D.A. 9.0.6		Id: 6/6